Giridhar Patri

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Education

University of Illinois at Chicago — Chicago, IL
Bachelor of Science in Computer Science, Minor in Mathematics — *Aug. 2021 – May 2025*

Technical Skills

Languages: C / C++, Java, Python, C#, F#,Typescript, Scala, SQL (Postgres), JavaScript, HTML/CSS, Rust
Frameworks: CUDA, React, Node.js, Qt, Flask, Spring-boot, Ruby on Rails, FastAPI, gRPC, PyTorch, Scikit-learn
Tools & CI/CD: Git, Github, Jira, Postman, Kubernates, Docker
Cloud Services: Firebase, DynamoDB, AWS EC2, AWS S3, AWS Lambda, Google Colab
Libraries: pandas, NumPy, Matplotlib, OpenCV, Playwright, Selenium

Professional Experience

Software Engineer — Oct. 2025 – Present
Visa

- Engineered a Python-based Security Assessment Orchestrator with Google ADK, Agent2Agent (A2A), and HTTP APIs to coordinate threat-modeling, penetration-testing, and remediation agents across vulnerability discovery, triage, patch generation, and fix validation
- Contributed to the open-sourcing of the platform as the Visa Vulnerability Agentic Harness (VVAH), translating Visa's Project Glasswing learnings into a reusable, multi-model framework for autonomous vulnerability discovery, remediation, and validation
- Built and launched a full-stack web application and MCP server integrating Claude Code with internal documentation, GitHub, Jira, and Confluence to embed project-aware lessons and dynamic coding challenges directly into AI-assisted development workflows

Software Engineering Intern — May. 2024 – Aug. 2024
Cleveland Cliffs

- Developed a full-stack React + .NET data visualization web app for telemetry and real-time updates using Axios API, Dapper ORM, and C#, coalescing a suite of outdated .ASP single-page applications for internal company engineers
- Built a steel coil tracking app with Qt framework & C++ backend, and added interaction with jQuery API improving inventory management, traceability, performance, and safety
- Led transformation of a locally-run Java application into web-based solution using Spring boot-framework, enabling remote access and increased scalability across the steel plant
- Ported legacy FORTRAN control modules to more modern C++20, contributing to long-term effort to modernize applications that interact with large-scale industrial machinery.

Machine Learning Research Assistant — Jan. 2023 – Dec. 2024
Mathematics & Computer Science Research Lab

- Implemented a neural-network chess engine in PyTorch, leveraging the LCZero open-source codebase, to explore novel reinforcement-learning–based evaluation functions and search strategies
- Created python-chess evaluation tool with the Lichess API to study how monotonicity and other heuristics impact engine strength
- Developed a testing environment/pipeline that trained and tested the CNN model over thousands of PGN files (plain-text game notation format that records gameplay) with Google's TPUs

Leadership

Quantitative Trading Club | *Executive Board Member* — Jan. 2022 – Dec. 2024

- Organized and competed in multiple trading competitions, across 20 campuses
- Lead a team of students to develop a custom trading API utilizing QuantConnect, a website to test trading algorithms
- Used basic trading algorithms/strategies like arbitrage algorithms, mean reversion, sentiment analysis (utilizing in-house QuantConnect API and Python) in various trading competitions with other top schools

Projects

Meet App | *Typescript, React Native, Redis, Firebase* — Aug. 2023 – Present

- Designed and built a location-based mobile social networking app using React Native (Expo) and Firebase
- Streamlines organization of large-scale meetups, and user interface for displaying meetup information and attendee location on an interactive map.